explore_energy_consumption

May 7, 2026

1 Time Series Energy Consumption

Using Eastern Interconnection grid data

[46]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import xgboost as xgb
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
import joblib
# from sklearn.impute import SimpleImputer
# from sklearn.preprocessing import PolynomialFeatures


color_pal = sns.color_palette()
plt.style.use('fivethirtyeight')
```

1.1 Introduction

This project evaluates time-series forecasting methods for energy consumption on the PJM Eastern Interconnection grid. While the initial framework followed the methodology by Rob Mulla, I significantly extended the scope through advanced feature engineering—specifically the integration of multi-horizon temporal lags (24-hour, 7-day, and multi-year). To optimize model performance, I implemented a robust hyperparameter tuning pipeline using GridSearchCV, executed on a private headless Linux server managed via Slurm and custom Bash automation.

1.1.1 Results

Model	RMSE	Improvement from Baseline	Notes
Baseline (Mean)	6,463.99	NA	Naive horizontal predictor
Linear Regression	5,342.17	17%	Base trend detection
XGBoost (Tutorial)	3,727.56	42%	Standard feature set (Rob Mulla)
Polynomial Regression	3,702.46	43%	Unstable (Degree 6, L2 alpha=1000)
XGBoost (Optimized)	**2,397.94**	**63%**	**Best Performance** (24h/7d Lags)

1.2 Discussion of Trade-offs

1.2.1 The "Wall" of Forecasting

While the **Lag-Optimized XGBoost** achieved the highest accuracy, it is bound by the **24-hour horizon**. Because the model relies on a 24-hour lag, predicting beyond that window requires a **Recursive Forecasting** strategy (feeding predictions back as inputs). This is powerful but sensitive to "drift" over long horizons.

1.2.2 Stability vs. Complexity

The **Polynomial Regression** (Degree 6) was theoretically competitive but mathematically unstable, as evidenced by the `LinAlgWarnings` during training. To stabilize it, a massive L2 penalty alpha=1000 was required. For real-world infrastructure like the PJM grid, the tree-based **XGBoost** model remains the superior choice due to its inherent stability and robustness against outliers.

[47]:
```
df = pd.read_csv('data/PJME_hourly.csv')
df = df.set_index('Datetime')
df.index = pd.to_datetime(df.index)

df.head()
```

[47]:

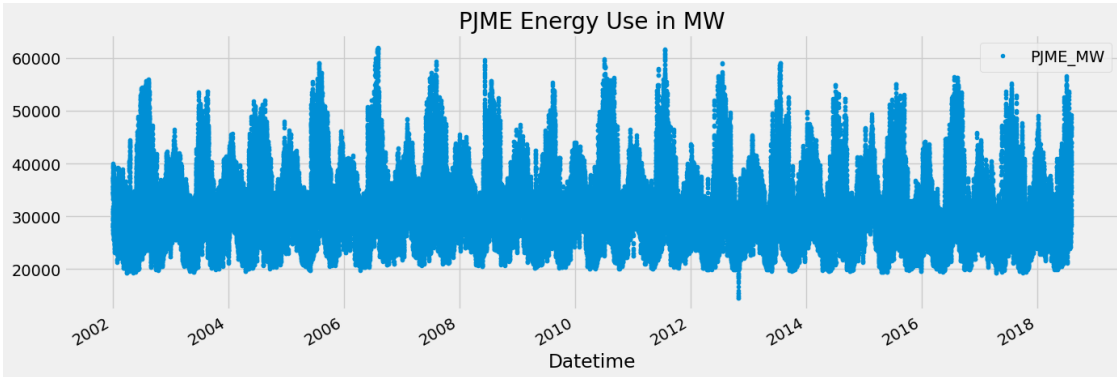
	PJME_MW
Datetime	
2002-12-31 01:00:00	26498.0
2002-12-31 02:00:00	25147.0
2002-12-31 03:00:00	24574.0
2002-12-31 04:00:00	24393.0
2002-12-31 05:00:00	24860.0

[48]:
```
target = df['PJME_MW']

mean_val = target.mean()
base_mse = ((target - mean_val)**2).mean()
base_rmse = (base_mse**0.5)

print(f'Mean: {mean_val:.2f}')
print(f'Baseline MSE: {base_mse:.2f}')
print(f'Baseline RMSE: {base_rmse:.4f}')
```

```
Mean: 32080.22
Baseline MSE: 41783165.85
Baseline RMSE: 6463.9899
```

[49]:
```
df.plot(style='.',
        figsize=(15,5),
        color=color_pal[0],
        title= 'PJME Energy Use in MW')
plt.plot
```

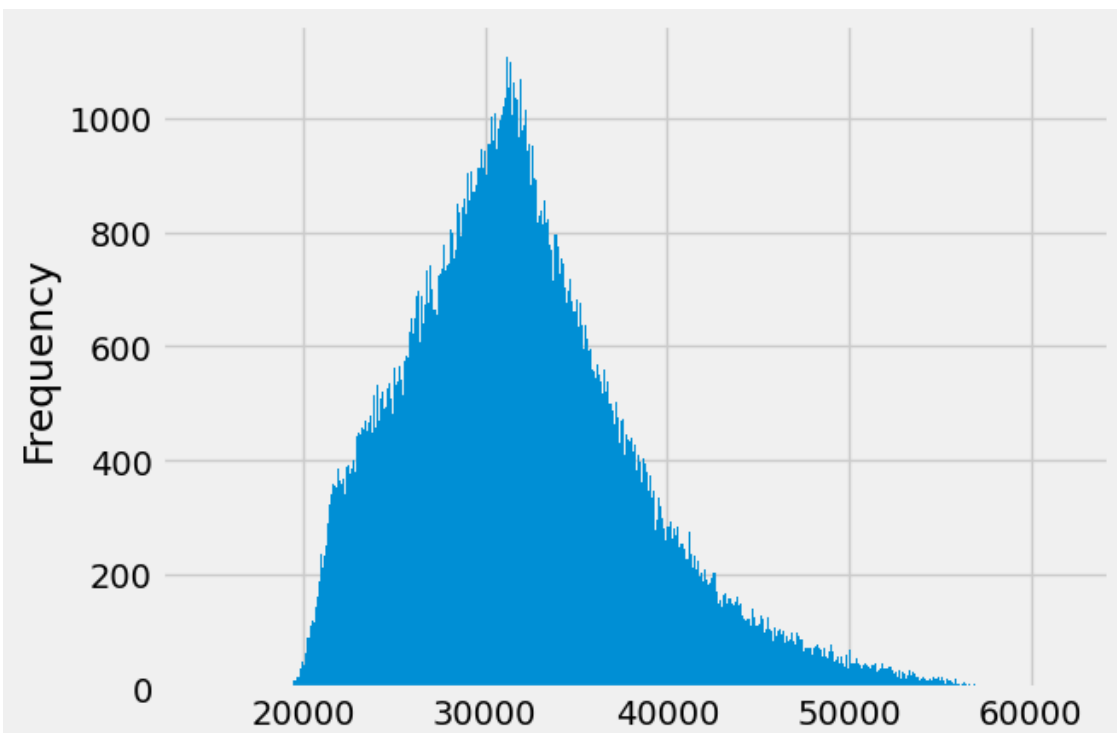
[49]: <function matplotlib.pyplot.plot(*args: 'float | ArrayLike | str', scalex: 'bool' = True, scaley: 'bool' = True, data=None, **kwargs) -> 'list[Line2D]'>



1.2.3 1. Outlier Analysis and Removal

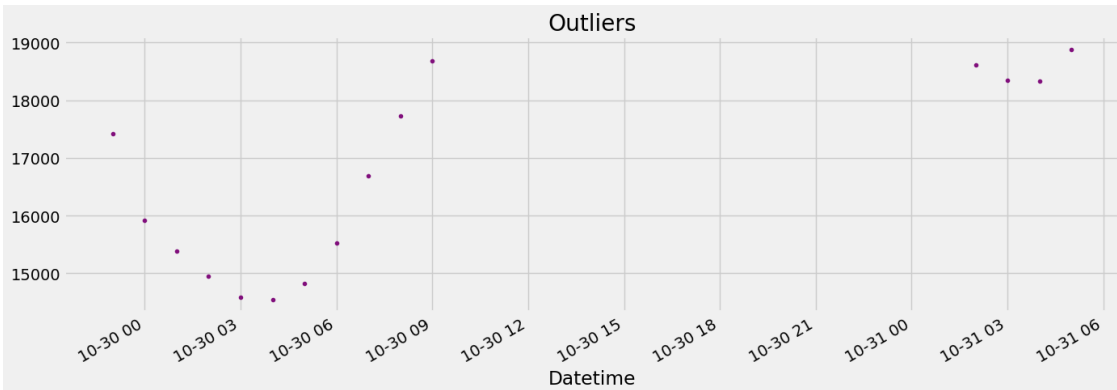
[50]:
```
df['PJME_MW'].plot(kind='hist', bins = 500)
```

[50]: <Axes: ylabel='Frequency'>



[51]:
```
df.query('PJME_MW < 19000')['PJME_MW'] \
    .plot(style='.',
        figsize=(15,5),
        color=color_pal[5],
        title= 'Outliers')
```

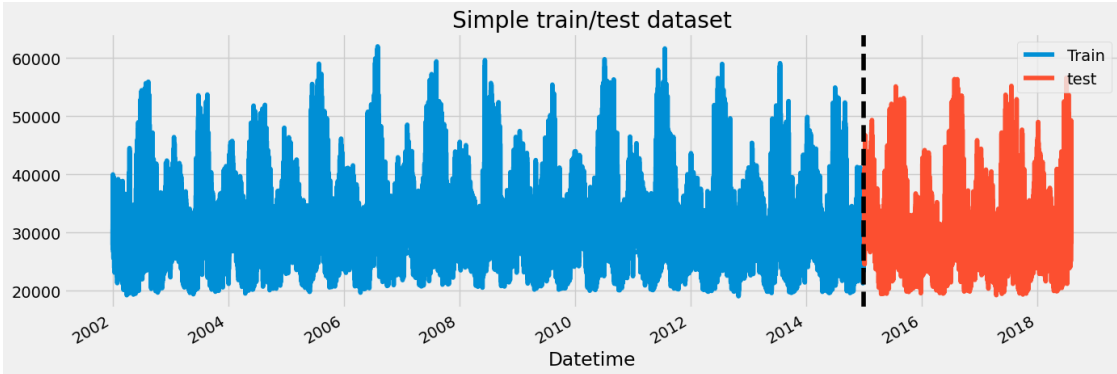
[51]: <Axes: title={'center': 'Outliers'}, xlabel='Datetime'>



[52]:
```
df = df.query('PJME_MW > 19000').copy()
```

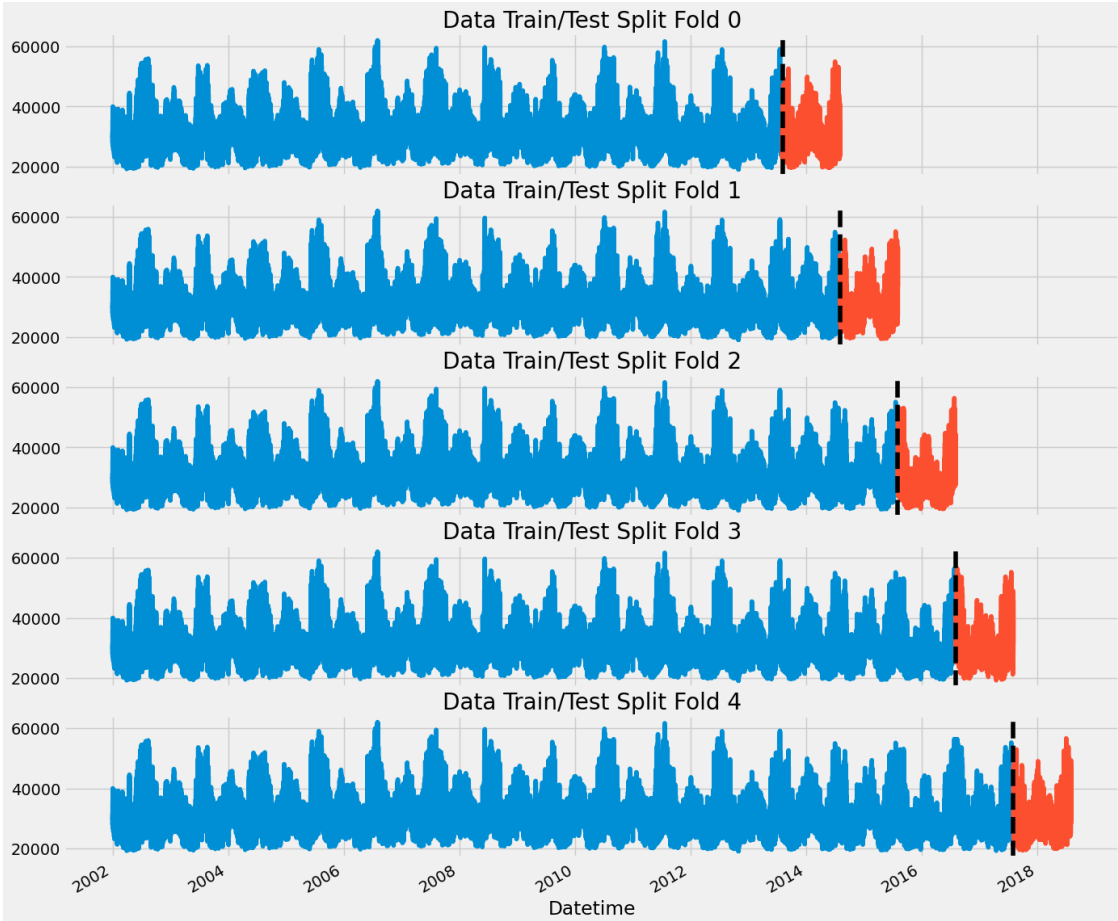
1.2.4 Simple train/test split

[53]:
```
train = df.loc[df.index < '01-01-2015']
test = df.loc[df.index >= '01-01-2015']

fig, ax = plt.subplots(figsize=(15,5))
train.plot(ax=ax, label='Training Set', title="Simple train/test dataset")
test.plot(ax=ax, label='Testing Set')
ax.axvline('01-01-2015', color='black', ls='--')
ax.legend(['Train', 'test'])

plt.show()
```



[54]:
```
from sklearn.model_selection import TimeSeriesSplit

tss = TimeSeriesSplit(n_splits=5, test_size=24*365*1, gap=24)
df = df.sort_index()
```

[55]:
```
fig, axs = plt.subplots(5, 1, figsize = (15, 15), sharex=True)

fold = 0
for train_indices, test_indices in tss.split(df):
    train = df.iloc[train_indices]
    test = df.iloc[test_indices]
    # print("first indices:")
    # print(test_indices[0])
    # print(train_indices[0])
    train['PJME_MW'].plot(ax=axs[fold],
                          label='Training Set',
                          title=f'Data Train/Test Split Fold {fold}')
    test['PJME_MW'].plot(ax=axs[fold],
                         label='Test Set')
    axs[fold].axvline(test.index.min(), color='black', ls='--')
    fold += 1
plt.show()
```



[56]:
```
import holidays

us_holidays = holidays.US()

def create_features(df):
    df = df.copy()
    # df['hour'] = df.index.hour
    df['hour_sin'] = np.sin(2 * np.pi * df.index.hour / 24)
    df['hour_cos'] = np.cos(2 * np.pi * df.index.hour / 24)
    df['dayofweek'] = df.index.dayofweek
    df['month'] = df.index.month
    df['year'] = df.index.year
    df['julianday'] = df.index.dayofyear
    df['weekofyear'] = df.index.isocalendar().week
    df['isholiday'] = [1 if d in us_holidays else 0 for d in df.index.date]

    return df
```

```
df = create_features(df)
```

[57]:
```
df.iloc[100]
```

[57]:
```
PJME_MW        26822.0
hour_sin      0.965926
hour_cos      0.258819
dayofweek          5.0
month              1.0
year            2002.0
julianday          5.0
weekofyear         1.0
isholiday          0.0
Name: 2002-01-05 05:00:00, dtype: Float64
```

[58]:
```
def add_lags(df):
    target_map = df['PJME_MW'].to_dict()
    df['lag1'] = (df.index - pd.Timedelta('364 days')).map(target_map)
    df['lag2'] = (df.index - pd.Timedelta('728 days')).map(target_map)
    df['lag3'] = (df.index - pd.Timedelta('1092 days')).map(target_map)
    return df

df = add_lags(df)
```

[59]:
```
df.columns
```

[59]:
```
Index(['PJME_MW', 'hour_sin', 'hour_cos', 'dayofweek', 'month', 'year',
       'julianday', 'weekofyear', 'isholiday', 'lag1', 'lag2', 'lag3'],
      dtype='object')
```

[60]:
```
fold = 0
preds = []
scores = []
for train_idx, val_idx in tss.split(df):
    print(f'--------------- fold: {fold} -------------')
    train = df.iloc[train_idx]
    test = df.iloc[val_idx]

    # train = create_features(train)
    # test = create_features(test)

    FEATURES = ['hour_sin', 'hour_cos', 'dayofweek', 'month', 'year',
       'julianday', 'weekofyear', 'isholiday', 'lag1', 'lag2', 'lag3']
    TARGET = 'PJME_MW'

    X_train = train[FEATURES]
    y_train = train[TARGET]
```

```
    X_test = test[FEATURES]
    y_test = test[TARGET]

    reg = xgb.XGBRegressor(base_score=0.5, booster='gbtree',
                           n_estimators=1000,
                           early_stopping_rounds=50,
                           objective='reg:squarederror',
                           max_depth=3,
                           learning_rate=0.01)
    reg.fit(X_train, y_train,
            eval_set=[(X_train, y_train), (X_test, y_test)],
            verbose=100)

    y_pred = reg.predict(X_test)
    preds.append(y_pred)
    score = np.sqrt(mean_squared_error(y_test, y_pred))
    scores.append(score)
    fold += 1
```

```
--------------- fold: 0 -------------
[0]     validation_0-rmse:32732.64189   validation_1-rmse:31956.84085
[100]   validation_0-rmse:12553.46393   validation_1-rmse:11934.91172
[200]   validation_0-rmse:5781.57545    validation_1-rmse:5393.57090
[300]   validation_0-rmse:3888.97186    validation_1-rmse:3903.75198
[400]   validation_0-rmse:3462.31548    validation_1-rmse:3761.15855
[459]   validation_0-rmse:3378.77389    validation_1-rmse:3762.49108
--------------- fold: 1 -------------
[0]     validation_0-rmse:32672.30157   validation_1-rmse:32139.24536
[100]   validation_0-rmse:12530.75691   validation_1-rmse:12252.96466
[200]   validation_0-rmse:5787.40648    validation_1-rmse:5690.48976
[300]   validation_0-rmse:3922.79581    validation_1-rmse:3924.26368
[400]   validation_0-rmse:3500.31533    validation_1-rmse:3594.34567
[500]   validation_0-rmse:3371.52509    validation_1-rmse:3486.00648
[600]   validation_0-rmse:3305.86173    validation_1-rmse:3431.87096
[700]   validation_0-rmse:3262.17061    validation_1-rmse:3404.49838
[800]   validation_0-rmse:3228.25502    validation_1-rmse:3383.48214
[900]   validation_0-rmse:3196.84830    validation_1-rmse:3368.57201
[999]   validation_0-rmse:3167.54114    validation_1-rmse:3365.46009
--------------- fold: 2 -------------
[0]     validation_0-rmse:32631.33667   validation_1-rmse:31073.73308
[100]   validation_0-rmse:12512.35445   validation_1-rmse:11165.37936
[200]   validation_0-rmse:5781.46173    validation_1-rmse:4802.84599
[300]   validation_0-rmse:3934.95418    validation_1-rmse:3522.04839
[400]   validation_0-rmse:3516.50083    validation_1-rmse:3502.48867
[404]   validation_0-rmse:3508.90304    validation_1-rmse:3505.44522
--------------- fold: 3 -------------
[0]     validation_0-rmse:32528.56343   validation_1-rmse:31477.56227
```

```
[100]	validation_0-rmse:12475.10987	validation_1-rmse:12057.56368
[200]	validation_0-rmse:5769.57520	validation_1-rmse:5813.44361
[300]	validation_0-rmse:3932.56121	validation_1-rmse:4371.91601
[400]	validation_0-rmse:3520.79405	validation_1-rmse:4175.61276
[500]	validation_0-rmse:3401.84664	validation_1-rmse:4133.92111
[600]	validation_0-rmse:3338.86460	validation_1-rmse:4092.68062
[700]	validation_0-rmse:3289.99020	validation_1-rmse:4065.02610
[800]	validation_0-rmse:3253.79290	validation_1-rmse:4060.16596
[821]	validation_0-rmse:3248.30785	validation_1-rmse:4060.93624
--------------- fold: 4 -------------
[0]	validation_0-rmse:32462.17074	validation_1-rmse:31465.62843
[100]	validation_0-rmse:12459.01963	validation_1-rmse:12000.70946
[200]	validation_0-rmse:5781.06671	validation_1-rmse:5637.47850
[300]	validation_0-rmse:3959.63363	validation_1-rmse:4154.44259
[400]	validation_0-rmse:3560.01202	validation_1-rmse:3995.11293
[488]	validation_0-rmse:3451.59276	validation_1-rmse:3989.54719
```

[61]:
```
print(f'Score across folds {np.mean(scores):0.4f}')
print(f'Fold scores:{scores}')
```

```
Score across folds 3727.5576
Fold scores:[np.float64(3758.848590474023), np.float64(3363.8107871944),
np.float64(3471.493362752243), np.float64(4054.581678725985),
np.float64(3989.053581622312)]
```

1.2.5 Predictioning with all data

[62]:
```
# df.tail()

FEATURES = ['hour_sin', 'hour_cos', 'dayofweek', 'month', 'year',
    'julianday', 'weekofyear', 'isholiday', 'lag1', 'lag2', 'lag3']
TARGET = 'PJME_MW'

X_all= train[FEATURES]
y_all = train[TARGET]

full_reg = xgb.XGBRegressor(base_score=0.5, booster='gbtree',
                            n_estimators=500,
                            objective='reg:squarederror',
                            max_depth=3,
                            learning_rate=0.01) \
                                .fit(X_all, y_all,
                                eval_set=[(X_all, y_all)],
                                verbose=500)
```

```
[0]	validation_0-rmse:32462.17074
[499]	validation_0-rmse:3443.60216
```

[63]:
```
df.index.max()
```

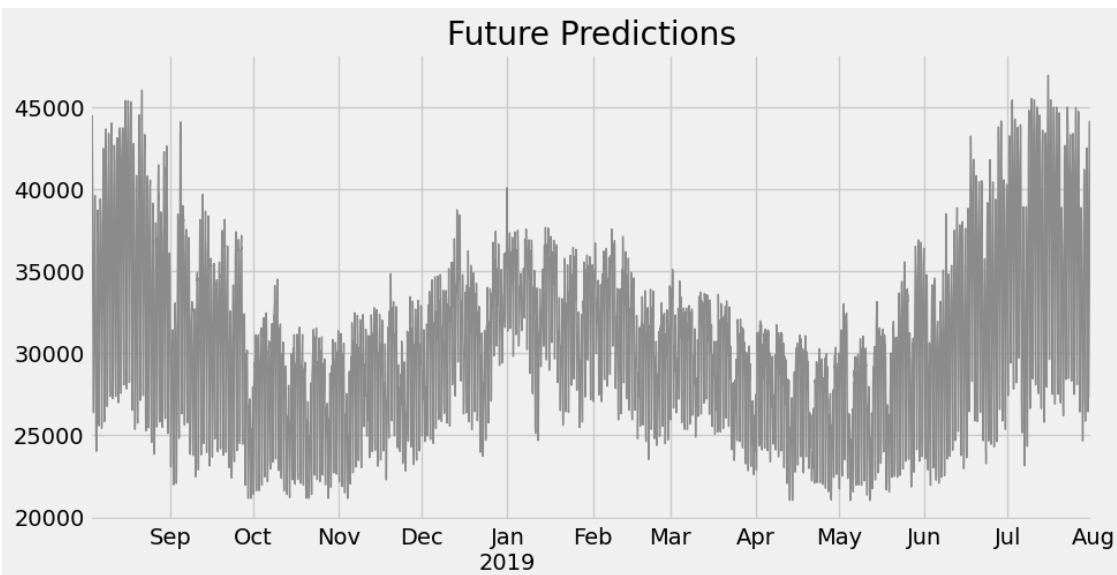
[63]: Timestamp('2018-08-03 00:00:00')

[64]:
```
future = pd.date_range('2018-08-03', '2019-08-01', freq='1h')
future_df = pd.DataFrame(index=future)
future_df['isFuture'] = True
df['isFuture'] = False
df_and_future = pd.concat([df, future_df])
df_and_future = create_features(df_and_future)
df_and_future = add_lags(df_and_future)
```

[65]:
```
future_w_features = df_and_future.query('isFuture').copy()
```

[66]:
```
future_w_features['pred'] = reg.predict(future_w_features[FEATURES])
```

[67]:
```
future_w_features['pred'].plot(figsize=(10, 5),
                               color=color_pal[4],
                               ms=1,
                               lw=1,
                               title='Future Predictions')
plt.show()
```



1.3 Linear Regression

[68]:
```
fold = 0
lin_preds = []
lin_scores = []
for train_idx, val_idx in tss.split(df):
   #  print(f'--------------- fold: {fold} -------------')
    train = df.iloc[train_idx]
    test = df.iloc[val_idx]

    # train = create_features(train)
    # test = create_features(test)

    FEATURES = ['hour_sin', 'hour_cos', 'dayofweek', 'month', 'year',
       'julianday', 'weekofyear', 'isholiday']
    TARGET = 'PJME_MW'

    X_train = train[FEATURES]
    y_train = train[TARGET]

    X_test = test[FEATURES]
    y_test = test[TARGET]

    lin_reg = LinearRegression()
    lin_reg.fit(X_train, y_train)

    y_pred = lin_reg.predict(X_test)
    lin_preds.append(y_pred)
    lin_score = np.sqrt(mean_squared_error(y_test, y_pred))
    lin_scores.append(lin_score)
    fold += 1

print(lin_preds)
print(np.mean(lin_scores))
```

```
[array([29912.47140393, 28031.67594414, 26953.4927012 , …,
       30622.44257054, 29289.91655071, 31884.08057421], shape=(8760,)),
array([30605.17978861, 29777.12137586, 29245.98177407, …,
       34212.85369098, 32893.21661812, 31015.26421779], shape=(8760,)),
array([29799.59067754, 28975.15500878, 28445.38585963, …,
       33361.08399004, 32049.82470861, 30186.11284128], shape=(8760,)),
array([28772.72678684, 27949.17675152, 27418.74820114, …,
       32293.03941721, 30986.54504068, 29129.91910331], shape=(8760,)),
array([27975.16314789, 27151.86989445, 26619.91188499, …,
       31476.62656933, 30175.16996164, 28334.08674579], shape=(8760,))]
5342.168834201302
```

2 24h Lag XGBoost Model

[69]:
```
# 1. Setup the Exact Features the Model Expects
FEATURES = ['hour_sin', 'hour_cos', 'dayofweek', 'month', 'year',
            'julianday', 'weekofyear', 'isholiday',
            'lag_24h', 'lag_7d', 'lag1', 'lag2', 'lag3']
TARGET = 'PJME_MW'

# 2. Load the trained model from the joblibs folder
daily_xgb_model = joblib.load('joblibs/PJME_hourly_model.joblib')
print("Model loaded successfully!")
```

```
Model loaded successfully!
```

[ ]:
```
def run_recursive_forecast(model, history_df, horizon_hours, target_col,␣
 ↪features_list):
    forecast_df = history_df.copy()
    future_predictions = []

    us_holidays = holidays.US()

    for _ in range(horizon_hours):

        next_time = forecast_df.index[-1] + pd.Timedelta(hours=1)

        next_row = pd.DataFrame(index=[next_time])

        next_row['hour_sin'] = np.sin(2 * np.pi * next_time.hour / 24)
        next_row['hour_cos'] = np.cos(2 * np.pi * next_time.hour / 24)
        next_row['dayofweek'] = next_time.dayofweek
        next_row['month'] = next_time.month
        next_row['year'] = next_time.year
        next_row['julianday'] = next_time.dayofyear
        next_row['weekofyear'] = next_time.isocalendar().week

        next_row['lag_24h'] = forecast_df.iloc[-24][target_col]
        next_row['lag_7d']  = forecast_df.iloc[-168][target_col]
        next_row['lag1']    = forecast_df.iloc[-8736][target_col]
        next_row['lag2']    = forecast_df.iloc[-17472][target_col]
        next_row['lag3']    = forecast_df.iloc[-26208][target_col]

        next_row['isholiday'] = 1 if next_time.date() in us_holidays else 0

        X_input = next_row[features_list]
        prediction = model.predict(X_input)[0]

        next_row[target_col] = prediction
```

```
        forecast_df = pd.concat([forecast_df, next_row])

        future_predictions.append({
            'Datetime': next_time,
            'Prediction': prediction
        })

    return pd.DataFrame(future_predictions).set_index('Datetime')
```

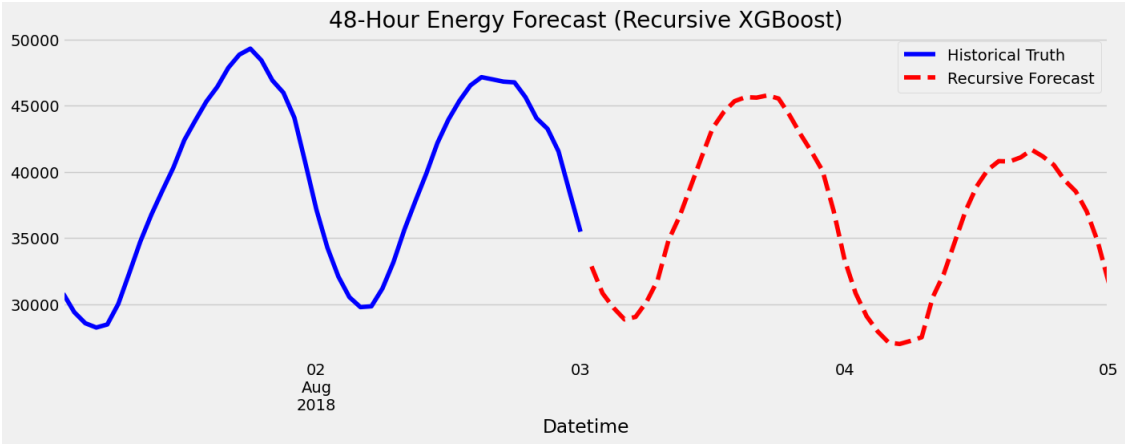
[ ]:
```
final_forecast = run_recursive_forecast(
    model=daily_xgb_model,
    history_df=df,
    horizon_hours=48,
    target_col=TARGET,
    features_list=FEATURES
)

final_forecast.head(10)
```

[ ]:
```
                      Prediction
Datetime
2018-08-03 01:00:00  32866.792969
2018-08-03 02:00:00  30839.277344
2018-08-03 03:00:00  29735.970703
2018-08-03 04:00:00  28863.375000
2018-08-03 05:00:00  29051.949219
2018-08-03 06:00:00  30189.255859
2018-08-03 07:00:00  31810.812500
2018-08-03 08:00:00  34842.804688
2018-08-03 09:00:00  36642.199219
2018-08-03 10:00:00  38878.066406
```

2.1 Plot 48 Hour Forecast

This is a recursive forecast. I realize that it can create exponiential errors, but I was more interested in the methods and how it worked.

[ ]:
```
fig, ax = plt.subplots(figsize=(15, 5))

df[TARGET].iloc[-48:].plot(ax=ax, label='Historical Truth', color='blue')

final_forecast['Prediction'].plot(ax=ax, label='Recursive Forecast',␣
 ↪color='red', linestyle='--')

ax.set_title("48-Hour Energy Forecast (Recursive XGBoost)")
ax.legend()
plt.show()
```


48-Hour Energy Forecast (Recursive XGBoost)
Historical Truth
Recursive Forecast
50000
45000
40000
35000
30000
02
Aug
2018
03
04
05
Datetime